Re-implementing *PatchTST* and Applying It to Retail Demand Forecasting

Final Project for CS 5782: Intro to Deep Learning (2026 Spring)

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Introduction

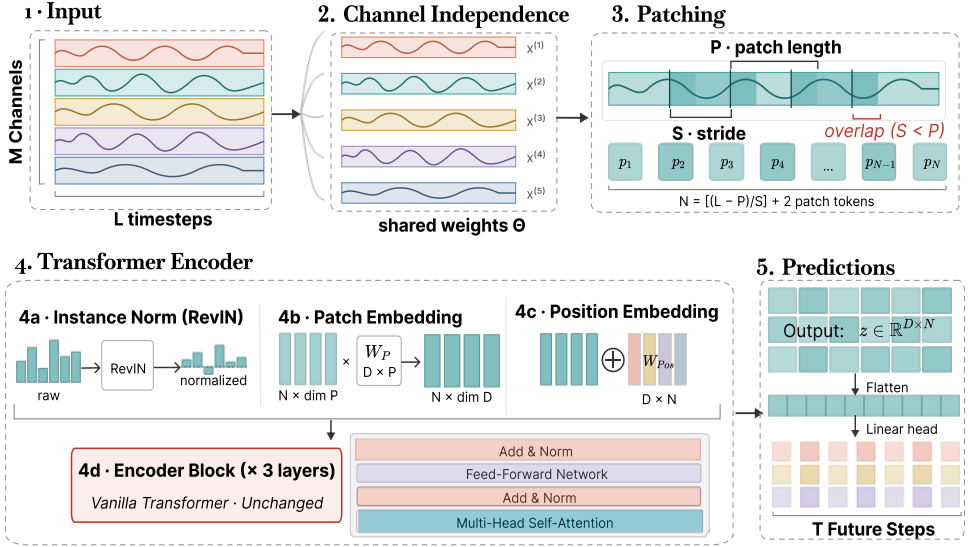
Motivation

- Transformers work well in NLP/CV
- **But underperform in long-term time series forecasting**
- Simple linear models (DLinear) can be surprisingly strong

Problem

- Existing Transformers use point-wise tokenization: lacking semantic information, $O(L^2)$ complexity
- Can patching make it useful for time-series (without changing architecture)?

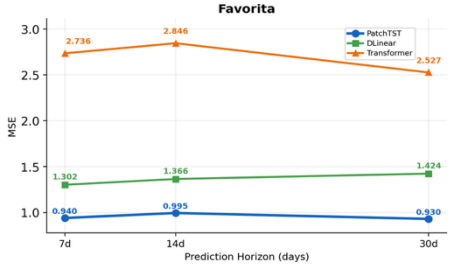
Methodology: Channel Independence + Patching + Transformer

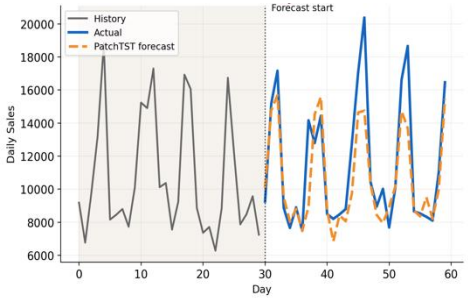


Application: Marketing Extension

Accurate demand forecasting for Favorita grocery sales data

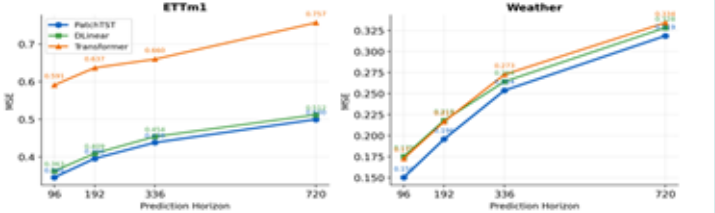
- Ecuador's largest supermarket chain
- Daily sales across 33 product categories (Grocery, Beverages, Dairy, etc.), 2012–2017
- Single store, categories as independent channels, L=336, horizons = 7d / 14d / 30d



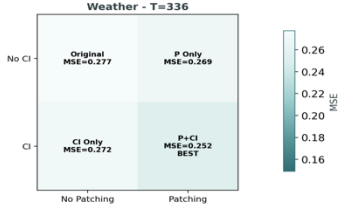


- CI is a natural fit : different product categories naturally have different demand drivers

Baseline: PatchTST vs DLinear vs Transformer



Effect of Patching and CI



Follow Up Work

- Cross-channel attention (iTransformer, ICLR 2024)
- Long context forecasting (Timer XL, ICLR 2025)
- Flexible patch (DeformableTST, NeurIPS 2024)

Conclusion: PatchTST's patching and channel independent design reproduce strong benchmark performance on forecasting time series and transfer well to retail demand forecasting.

References: Nie et al.,"A Time Series is Worth 64 Words: Long-term Forecasting with Transformers," ICLR 2023.